

Article

Assessing the Individual and Combined Contributions of Stand Age and Tree Height for Regional-Scale Aboveground Biomass Estimation in Fast-Growing Plantations

Xiaomin Li ^{1,2} , Dan Zhao ^{1,2,3,*} , Junhua Chen ^{1,2}, Jinchun Wu ^{1,2}, Xuan Mu ^{1,2}, Zhaoju Zheng ¹, Cong Xu ¹, Chunjie Fan ^{4,5} , Yuan Zeng ^{1,2} and Bingfang Wu ^{1,2,3} 

¹ State Key Laboratory of Remote Sensing and Digital Earth, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China; lixiaomin231@mails.ucas.ac.cn (X.L.); chenjunhua21@mails.ucas.ac.cn (J.C.); wujinchen22@mails.ucas.ac.cn (J.W.); muxuan22@mails.ucas.ac.cn (X.M.); zhengzhaoju@aircas.ac.cn (Z.Z.); xucong@aircas.ac.cn (C.X.); zengyuan@aircas.ac.cn (Y.Z.); wubf@radi.ac.cn (B.W.)

² University of Chinese Academy of Sciences, Beijing 100049, China

³ Beijing Yanshan Earth Critical Zone National Research Station, University of Chinese Academy of Sciences, Beijing 101408, China

⁴ State Key Laboratory of Tree Genetics and Breeding, Key Laboratory of State Forestry and Grassland Administration on Tropical Forestry, Research Institute of Tropical Forestry, Chinese Academy of Forestry, Guangzhou 510520, China; fanchunjie@caf.ac.cn

⁵ State Key Laboratory of Subtropical Silviculture, College of Forestry and Biotechnology, Zhejiang A&F University, Hangzhou 311300, China

* Correspondence: zhaodan@aircas.ac.cn; Tel.: +86-186-1162-2725

Abstract

Accurate estimation of plantation aboveground biomass (AGB) is critical for quantifying carbon cycles and informing sustainable forest resource management, but enhancing estimation accuracy remains a key challenge. Although tree height and stand age are recognized as critical predictors for enhancing AGB models in addition to spectral vegetation indices, their individual and combined contributions in regional plantation forests remain insufficiently quantified, especially concerning the potential for leveraging the distinct characteristics of fast-growing plantations to facilitate AGB estimation. This study developed multi-source remote sensing-based *Eucalyptus* AGB estimation models for Nanning, Guangxi, integrating stand age and tree height to assess their impacts. Stand age was mapped from Landsat time-series imagery, and tree height was derived from UAV-LiDAR data. Plot-level reference AGB was obtained using fused UAV and terrestrial LiDAR point clouds. A random forest model, incorporating these variables with Sentinel-2 spectral information and topography, then achieved regional AGB estimation. The findings demonstrate that (1) tree height serves as the most influential predictor for AGB estimation at the regional scale, yielding a robust model performance ($R^2 = 0.84$). (2) Tree height captures the majority of the explanatory power associated with stand age. Once tree height was included as a predictor, the subsequent addition of stand age offered no significant improvement in model accuracy ($R^2 = 0.85$). (3) Given the challenges in obtaining precise tree height data and the robust correlation between stand age and tree height in fast-growing plantations, the integration of stand age substantially improved the accuracy of AGB estimations (from the spectral model of $R^2 = 0.54$ to $R^2 = 0.74$), with performance approaching that of tree height-based models ($\Delta R^2 = 0.10$). Consequently, in fast-growing plantations, which are often characterized by high stand homogeneity, a hybrid model incorporating stand age can offer a reliable and cost-effective solution for AGB estimation.



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1. Introduction

Forests, as significant carbon sinks on land, play a crucial role in mitigating and adapting to climate change [1]. Aboveground biomass (AGB) serves as a key factor for assessing forest carbon stocks and carbon sink capacity and an important indicator for evaluating forest ecological functions and ecosystem health [2]. It is defined as the total amount of organic matter in the aboveground parts of vegetation (including trunks, branches, and leaves) per unit area over a specified time period [3]. Plantation forests contribute significantly to rapid carbon accumulation and the supply of timber for economic purposes [4]. In particular, fast-growing plantations (e.g., *Eucalyptus*, *Populus*) exhibit more pronounced carbon sequestration rates due to their accelerated growth cycles and shorter rotation periods compared to conventional plantations [5]. Accurate AGB estimation in plantation forests is therefore important for carbon accounting and provides a foundational understanding of anthropogenic influences on carbon cycling, informing global carbon budget models and nature-based climate solutions [6]. Remote sensing, offering large-scale and continuous observation capabilities, has emerged as a primary tool for AGB estimation; however, enhancing accuracy and spatiotemporal coverage remains challenging [7].

Traditional AGB estimation using remote sensing primarily relies on spectral information, vegetation indices, radar polarization characteristics, and topographic factors provided by multi-source remote sensing data. These methods typically employ statistical models or machine learning models to extrapolate from sample plots to larger scales [8,9]. However, optical remote sensing-based methods are usually constrained by uncertainty introduced by positional biases [10], limited ground plot data [11], and saturation effects in high-biomass regions [12].

Tree height, an important forest structural parameter, exhibits a significant allometric relationship with AGB. Incorporating tree height as a key variable into AGB estimation models has become a core research direction to improve estimation accuracy [13,14]. Light Detection and Ranging (LiDAR) technology, with its precise capability to capture vertical forest structures (such as tree height and canopy density) [15], is widely recognized as a key bridge between ground surveys and large-scale remote sensing estimates. Its application has not only enhanced the spatial representativeness and extrapolation ability of ground plot data but also effectively mitigated estimation errors caused by spectral saturation, positional biases, and sparse sampling plots [16]. However, while airborne LiDAR (e.g., unmanned aerial vehicle (UAV)) delivers high spatial precision, it is constrained by limited coverage and prohibitive costs, thereby hindering its utility for extensive, continuous dynamic monitoring [17]. Spaceborne LiDAR (e.g., GEDI), while offering certain advantages in coverage and data acquisition frequency, contends with challenges such as discrete sampling, low spatial resolution, limited vertical accuracy [18], and significant reductions in accuracy in areas with notable terrain variation [19]. Overall, tree height data can significantly enhance AGB estimation accuracy, but the acquisition of high-precision tree height data at the regional scale remains a formidable challenge, thus impeding its widespread implementation.

Stand age directly modulates the spectral response and canopy structure of vegetation through the growth of tree crowns and tree height. Incorporating stand age information into AGB remote sensing estimation models can bolster temporal adaptability and estimation accuracy [20,21]. Currently, stand age determination predominantly hinges on

disturbance detection in remote sensing time-series, where sudden changes in vegetation indices or spectral characteristics over time are identified using various algorithms to infer vegetation renewal or disturbance events, subsequently facilitating stand age estimation. However, current disturbance detection methods still face numerous challenges in large-scale applications in natural forests. These challenges include limited remote sensing time-series length, substantial inter-algorithm variability in disturbance detection accuracy (e.g., Landsat-based Detection of Trends in Disturbance and Recovery (LandTrendr) and Continuous Change Detection and Classification (CCDC)) in natural forests [22], and the inherent difficulty of detecting gradual disturbances, which can be obscured by noise [23]. Among these, despite potential minor omission errors at fine spatial scales [24], LandTrendr is broadly favored due to its capacity to detect both discrete and gradual events, low omission rates, and advantages in time-series modeling. In contrast, in plantation forests, especially fast-growing plantations managed with short and regular rotation cycles and characterized by uniform planting density, pronounced anthropogenic disturbances, and evident disturbance periodicity [25,26], disturbance signals are more easily identifiable [27]. Thus, disturbance detection is more applicable and reliable in these types of forest stands.

This study focuses on the central region of Nanning, where *Eucalyptus* AGB estimation models are developed by incorporating spectral, topographic, stand age, and tree height information, which aims to validate the optimization performance achieved by introducing stand age and tree height parameters. Based on this and considering the cost and availability limitations of high-precision tree height data, this study explores whether stand age can serve as an effective substitute variable for fast-growing plantations AGB modeling in scenarios where tree height observation is constrained.

2. Materials and Methods

2.1. Study Area

The study area is located in the central region of Nanning, Guangxi, spanning longitude 107°45' to 109°02'E and latitude 22°34' to 23°30'N, encompassing the largest state-owned Gaofeng Forest Farm in Guangxi, China (Figure 1). Climatically, the area is characterized by a humid subtropical monsoon regime, providing ample solar radiation and precipitation. Topographically, the landscape is predominantly hilly, exhibiting substantial elevational variability from approximately 100 m to 460 m above sea level, and slopes typically ranging between 6° and 35° [28]. The main tree species in the area include *Cunninghamia lanceolata*, *Pinus massoniana*, and *Eucalyptus grandis* × *urophylla* [29]. Crucially, as a primary hub for *Eucalyptus* cultivation and timber harvesting, Gaofeng Forest Farm is characterized by frequent and well-documented stand rotation activities. This consistent management regime consequently provides a robust and well-defined dataset, rendering the site exceptionally suitable for the objectives of this study.

2.2. Satellite Data

This study utilized Landsat 8 Collection 2 Level-2 surface reflectance products from the USGS Earth Explorer (Reston, VA, USA; <https://earthexplorer.usgs.gov/>, accessed on 11 August 2025), covering the forest growing season from June to September each year from 2013 to 2022. These data products have undergone standardized preprocessing, including systematic radiometric calibration, atmospheric correction, and geometric registration. Landsat 8 imagery was selected primarily for its long-term, consistent observational record, which is essential for detecting forest dynamics and inferring stand age. Cloud masking was implemented using embedded Quality Assessment (QA) bands, followed by annual cloud-free composite generation through median compositing algorithms. The Seamless Mosaic tool within the ENVI software (version 5.6; Broomfield, CO, USA) was then utilized

to generate mosaic imagery from adjacent orbital paths. Further preprocessing, including image clipping and reprojection, was conducted. These temporal image stacks and their associated spectral variations were subsequently analyzed to accurately infer forest stand age, with results resampled to a 10 m resolution to meet modeling precision requirements.

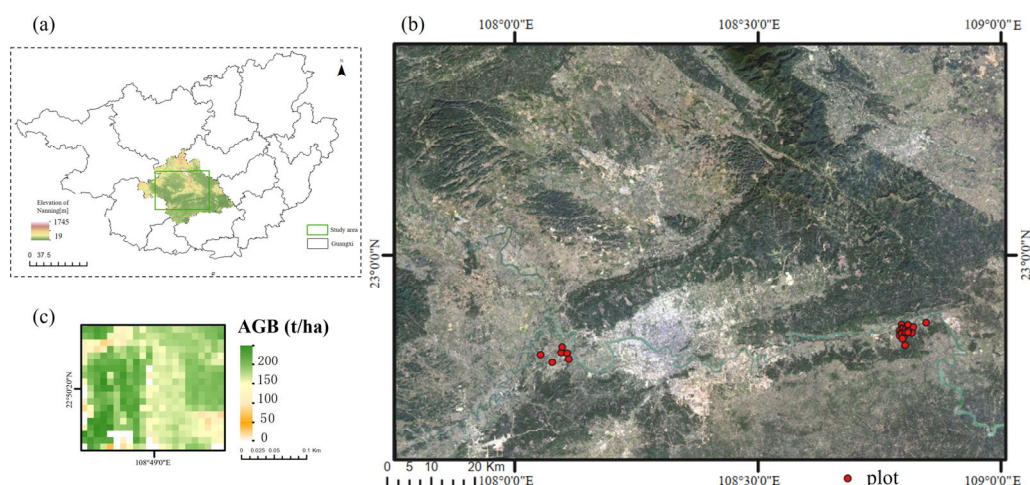


Figure 1. (a) Location of the study area. (b) True-color composite image (R: B4, G: B3, B: B2) of Landsat 8 Collection 2 Level-2 surface reflectance data for the study area and locations of sample plots. (c) An example of AGB estimation results at 10-m resolution plots.

To construct high-resolution AGB models, Sentinel-2 L2A data from October 2022, covering the Gaofeng Forest Farm region, were downloaded from the European Space Agency's Copernicus Data Centre (Paris, France; <https://dataspace.copernicus.eu/>, accessed on 11 August 2025) (Figure 1b). While the visible and near-infrared bands offered a 10 m spatial resolution, the shortwave infrared (SWIR) bands were at 20 m. Given the importance of SWIR in AGB estimation, those SWIR bands were resampled to a 10 m spatial resolution to ensure consistency with the AGB reference data. Following similar preprocessing steps of mosaicking and subsetting, their spectral information was extracted for a 10 m-resolution AGB estimation model tailored for fast-growing plantations. Furthermore, historical high-resolution imagery from Google Earth Pro (Mountain View, CA, USA) was additionally incorporated as supplementary validation material and ancillary reference data.

2.3. UAV LiDAR Point Cloud and AGB Data from Sample Plots

In our previous research, we acquired two types of LiDAR data between October and December 2022. Terrestrial LiDAR data were collected from 40 sample plots (20 m × 30 m each) using the UA Terrestrial Laser Scanning System (SureStar, Beijing, China), and UAV flight data were captured from 24 plots (200 m × 200 m each) with the Genius LiDAR sensor (SureStar, Beijing, China) mounted on a DJI M200 UAV (DJI, Shenzhen, China). Fused point clouds were generated via hierarchical clustering, feature extraction, and point cloud registration, followed by three-dimensional reconstruction of the fused point clouds. Then single-tree-level AGB estimation was subsequently performed utilizing the Quantitative Structure Model (QSM) and a volume density method [30]. The results demonstrated strong agreement (CCC = 0.93) between TreeQSM-based AGB estimates and those derived from locally calibrated allometric equations based on destructive sampling. The total AGB for each sample plot was subsequently aggregated through single-wood AGBs' summation across 10 × 10 m grid cells, establishing foundational biomass references for machine learning calibration. A series of parameters aimed at comprehensively capturing forest structural features were extracted from the UAV LiDAR point cloud data in the

corresponding sample plot areas. These parameters were then analyzed through random forest (RF) regression to establish a robust plot-scale AGB estimation model. Finally, this plot-scale AGB model was then applied to generate 10 m-resolution AGB data for the 24 reference plots (Figure 1c) [31]. Concurrently, fine-scale tree species' distribution within the study area was generated utilizing 2022 GF-2 (China Centre for Resources Satellite Data and Application, Beijing, China) satellite imagery and a U-Net semantic segmentation model. In this study, the growth range of *Eucalyptus* was delineated from this output to serve as the final result mask for the AGB mapping.

2.4. Auxiliary Data

The digital elevation model (DEM) employed in this study was sourced from the 30m-spatial-resolution Shuttle Radar Topography Mission (SRTM) data, disseminated by NASA (Washington, DC, USA). From this DEM, key topographic parameters, including slope, aspect, and the Topographic Solar Radiation Index (TSRI), were extracted and subsequently incorporated as explanatory variables in the *Eucalyptus* AGB estimation model. Additionally, to rigorously assess the interference results and characterize potential uncertainties for *Eucalyptus*, a comprehensive validation dataset was established. Due to the lack of direct and precise in situ forest age records within the study area, a proxy-based estimation method using high-resolution remote sensing imagery was adopted. Specifically, 300 sample points were randomly generated within the Gaofeng Forest Farm. We identified land cover changes by visually interpreting time-series imagery from Google Earth Pro, performing detailed image comparisons, neighborhood-based integrated analysis, and sequential time-series analysis to mitigate the inherent spatial uncertainty of the imagery. The analysis focused on pinpointing temporal markers such as deforestation, vegetation recovery, and new planting activities. By synthesizing the results from the multi-temporal image comparison with auxiliary information, including existing field survey data and expert knowledge, an approximate forest age was estimated for each sample point. Detailed validation results are presented in Figure 2 and Table 1.

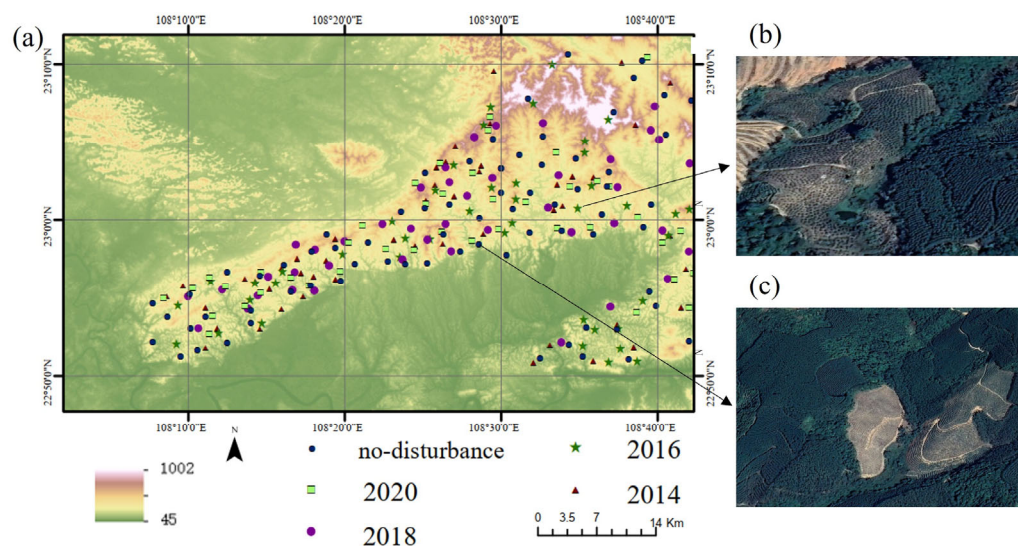


Figure 2. (a) Sample locations of the Gaofeng Forest Farm. (b,c) Google Earth imagery for validation.

Table 1. Sample data for plantation disturbance and stand age validation.

	2014	2016	2018	2020	Total
Disturbance Points	50	50	50	50	200
No-Disturbance Points	25	25	25	25	100

2.5. Methods

The overall methodology employed in this research is conceptually illustrated in Figure 3. Initially, leveraging Landsat time-series imagery and the LandTrendr sliding-time-window algorithm, forest disturbance events were identified within the study area, from which forest stand age was subsequently derived. Following this, an RF regression model was developed to estimate AGB by integrating stand age data obtained through disturbance time inversion, spectral and topographic features from multispectral satellite imagery, and canopy height data derived from UAV LiDAR point clouds. Notably, focusing on tree height, a key parameter for AGB estimation that is difficult to obtain accurately at large scales, this study emphasizes the significant correlation between stand age and tree height to evaluate the feasibility of substituting tree height with stand age data. Finally, region-scale AGB mapping for *Eucalyptus* forests was achieved based on the stand age-optical data model.

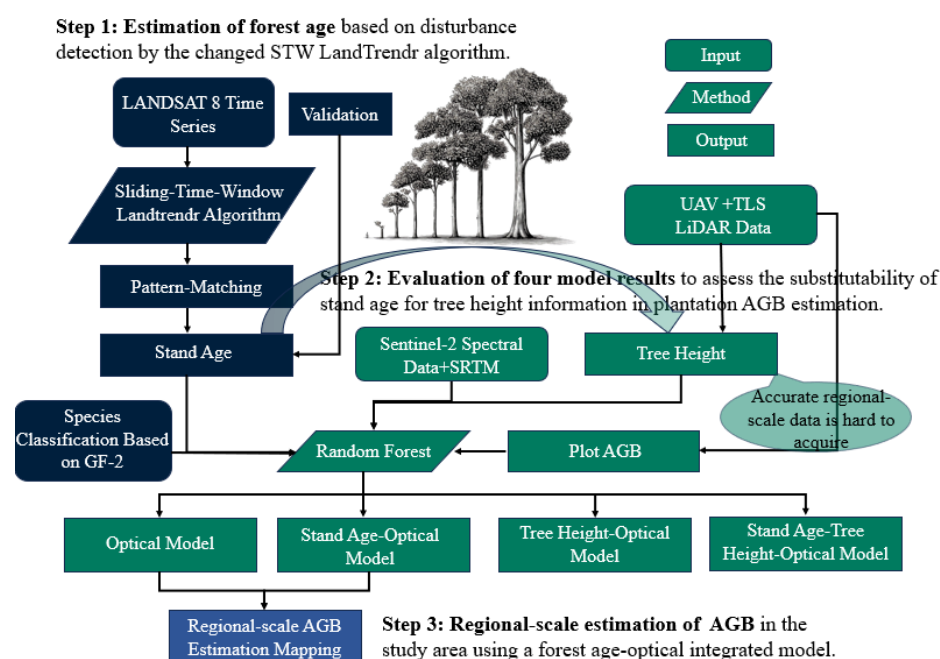


Figure 3. Method flowchart of this study.

2.5.1. Stand Age Monitoring Based on the LandTrendr Sliding-Time-Window Algorithm

The LandTrendr algorithm is a set of spectral temporal segmentation algorithms used to detect changes in medium-resolution satellite imagery time series (primarily from Landsat series images) and generate trajectory-based spectral time-series data that are largely free from inter-annual signal noise [32,33]. LandTrendr models change in the imagery time series using line segments, where each vertex of the line segment represents a change point, and the minimum time interval between vertices is one year. From the vertex information, the start year and magnitude of disturbance events can be determined [34]. Li et al. (2024) proposed a methodology grounded in the Sliding Time Window Change Detection (STWCD) algorithm [35], whereby monthly NDVI time series are constructed, and LandTrendr-derived change detections within each sliding window are integrated to extract and analyze the historical dynamics of *Eucalyptus* plantations in the Guangxi region. This study refers to this method and makes improvements based on the study area's conditions to extract the planting timeline of *Eucalyptus*.

Different spectral indices have varying capabilities in monitoring disturbance events. Through a preliminary evaluation of the algorithm's ability to characterize forest disturbance events in the study area using different vegetation indices, this study concludes that

the Normalized Burn Ratio (NBR) index offers superior monitoring efficiency than NDVI. Therefore, the NBR index is selected for monitoring disturbances in *Eucalyptus* plantations. The specific calculation formula is shown in Equation (1).

$$\text{NBR} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR}) \quad (1)$$

where NIR refers to the near-infrared band, and SWIR denotes the short-wave infrared band.

By evaluating the quality of imagery data from the study area, the annual NBR time series was selected for subsequent research. A sliding-time-window algorithm was applied to segment the NBR time series. Based on the biological characteristics of *Eucalyptus*, the rotation model, and experimental test results, the window length was set to 6 years with a step size of 1 year. The LandTrendr algorithm was used to calculate the NBR change characteristics for each time segment, identifying key time points for *Eucalyptus* planting or harvesting. The LandTrendr parameters were adjusted to achieve optimal curve fitting, with the final parameters provided in Table 2. Algorithm parameters were configured with reference to the original literature and locally optimized to accommodate the rapid growth characteristics of *Eucalyptus* plantations in the study area. Key parameters, such as window length and pattern matching threshold, were fine-tuned through iterative trials to maximize detection accuracy.

Table 2. Parameters of LandTrendr algorithm.

Parameter	Type	Values	Definition
maxSegments	Integer	20	Maximum number of segments to be fitted on the time series.
spikeThreshold	Float	0.9	Threshold for dampening the spikes (1.0 means no dampening).
vertexCountOvershoot	Integer	3	The initial model can overshoot the maxSegments + 1 vertices by this amount. Later, it will be pruned down to maxSegments + 1.
preventOne YearRecovery	Boolean	False	To prevent recovery events within the same year.
recovery Threshold	Float	0.9	If a segment has a recovery rate faster than $1/\text{recoveryThreshold}$ (in years), then the segment is disallowed.
pvalThreshold	Boolean	0.05	If the p -value of the fitted model exceeds this threshold, then the current model is discarded and another one is fitted using the Levenberg–Marquardt optimizer.
bestModelProportion	Float	0.5	Takes the model with the most vertices that has a p -value that is at most this proportion away from the model with lowest p -value.
minObservationsNeeded	Integer	6	Minimum observations needed to perform output fitting.

A pattern matching method was employed to filter the detection results and identify valid *Eucalyptus* planting events, with the identification conditions given in Equations (2)–(4). The sliding window detection was carried out iteratively, and the results from all windows were integrated at the pixel level using the maximum value to determine the most recent disturbance occurrence. Further post-processing, such as morphological operations, was applied to optimize the results.

$$\text{Mag} \geq 0.20 \quad (2)$$

$$20 \leq \text{Rate} \leq 200 \quad (3)$$

$$Preval \geq 0.55 \quad (4)$$

where *Mag* represents the magnitude of change, *Rate* indicates the rate of change, and *Preval* refers to the stable spectral index value before the change occurred.

Significant forest disturbance events, such as extensive harvesting operations and high-severity fires, result in the near-complete removal of aboveground biomass, compelling the ecosystem to initiate successional pathways [36]. In such instances, the temporal marker of disturbance occurrence serves as the definitive initiation point for stand age determination, with stand age consequently defined as the time elapsed since the cessation of the disturbance event and the onset of recovery [37]. Field surveys have revealed that *Eucalyptus* cultivation is primarily divided into two types: coppice management and replanted management. Coppice systems involve the retention of stumps post-harvest, facilitating subsequent regeneration from the extant rootstock, typically by allowing selected sprouts (one to three per stump) to develop. Conversely, replanted management is typically implemented when, following two or more cycles of sprout regeneration, diminished growth vigor and increased mortality rates become evident, leading to stand irregularities or necessitating replanting with new genetic material [38]. Due to the short rotation period of *Eucalyptus* plantations, most *Eucalyptus* stands within the same pixel underwent no more than two harvesting events between 2013 and 2022. The most recent disturbance prior to 2022 is considered as the stand age, with disturbances occurring in 2021 assigned a stand age of 1, and areas with no detected disturbances assigned the maximum stand age (9 years).

2.5.2. Regional-Scale AGB Modeling Based on Multispectral Satellite Data

In the acquired sample plot data, this study further optimized and filtered the plots, combining high-resolution remote sensing image interpretation, the DEM and its derived topographic factors (such as slope and aspect), as well as other auxiliary multi-source information. From this, plots unambiguously located within *Eucalyptus* plantation boundaries and representative of diverse site environments were systematically selected, ultimately identifying a total of 2595 distinct and representative *Eucalyptus* plantation sample points.

Tree height was derived from high-precision UAVLiDAR point cloud data. The data underwent a series of processing steps, including trajectory solving, point cloud calculation, and clipping/denoising, to ensure high geometric accuracy (point density > 200 pts/m²). The resulting tree height estimates typically have an error margin of 10–20 cm [39,40]. The UAV LiDAR point cloud data for the sample plots were used to extract the 95th percentile of point cloud height (H95) as a representative parameter for tree height. Combined with the historical images of Google Earth, the temporal analysis was carried out to remove the outliers in the sample AGB and tree height data through manual inspection, and the stand age data were verified and corrected to ensure robust data accuracy and reliability.

Logarithmic fitting analysis was performed to quantify the relationships between stand age and tree height and between stand age and AGB (Figure 4). The results showed that the relationship between tree height and stand age had a higher fitting accuracy ($a = 7.82$, $b = 3.00$, and $R^2 = 0.73$), while the relationship between AGB and stand age had relatively lower fitting accuracy ($a = 46.97$, $b = 50.03$, and $R^2 = 0.54$). During the development of the *Eucalyptus* stand age–tree height relationship model, a systematic underestimation of some observed values was identified. Further investigation attributed this discrepancy to a potential time lag exceeding six months between logging activities and subsequent replanting cycles, which implies that the observed stand age could be chronologically older than the actual physiological age of the current stand. Given the rapid early-stage growth of *Eucalyptus* (typically 2–3 m within a six-month period), this chronological offset likely contributed to the observed underestimations. Overall, the growth trend observed

in this study was similar to the findings of Chen et al. (2011) [41] regarding the growth patterns of *Eucalyptus grandis* in the Dongmen Forest Farm in Guangxi. Considering that *Eucalyptus* growth is greatly influenced by management practices and planting density [42], the tree height growth trend observed in this study aligns closely with the growth patterns of *Eucalyptus*. Furthermore, the good fit between stand age and tree height confirms the feasibility of using stand age as an effective surrogate for tree height data in regional-scale *Eucalyptus* plantation AGB estimation when measured tree height data is unavailable. While we acknowledge that the volume-based AGB estimation approach may partially contribute to the observed strong correlation between tree height and AGB, we maintain that the high homogeneity of fast-growing plantations is the dominant factor underlying this strong height–AGB relationship.

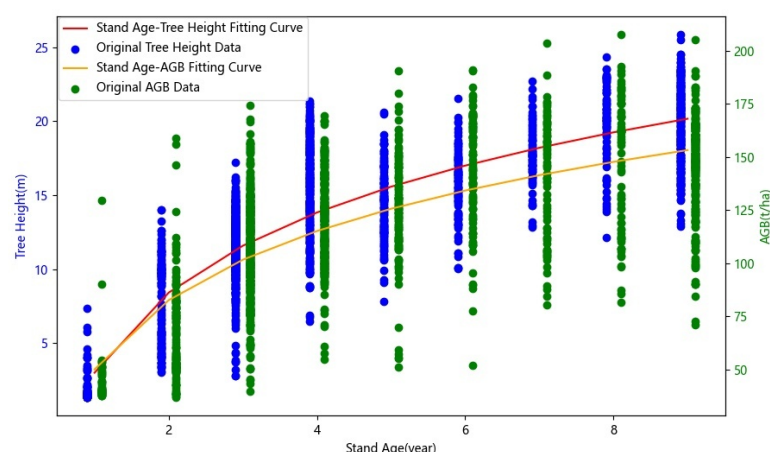


Figure 4. Logarithmic relationships between stand age and tree height as well as AGB.

Vegetation indices (VIs) are often computed using similar spectral bands, which can lead to significant multicollinearity. While RF possess some inherent tolerance to multicollinearity, excessive correlation can still impact the assessment of feature importance. In addition to selecting features based on their importance scores, we employed Variance Inflation Factor (VIF) analysis in preliminary experiments to identify and remove highly collinear indices such as NDRE and GDVI; ultimately, this study selected 10 vegetation indices [43,44] (Table 3), along with spectral bands from the blue, green, and red visible-light bands (Blue, Green, and Red), near-infrared (NIR) bands, and short-wave infrared bands (SWIR1; SWIR2) as input variables for constructing the AGB estimation model for the regional scale. Additionally, SRTM data were processed to obtain elevation, slope, aspect, and the TSRI, which were incorporated as important topographic variable.

Table 3. Common vegetation indices.

Vegetation Index	Calculation Formula
Normalized Difference Vegetation Index	$NDVI = (NIR - Red) / (NIR + Red)$
Enhanced Vegetation Index	$EVI = 2.5((NIR - Red) / (NIR + 6Red - 7.5Blue + 1))$
Normalized Difference Infrared Index	$NDII = (NIR - SWIR1) / (NIR + SWIR1)$
Soil Adjusted Vegetation Index	$SAVI = (1 + L)(NIR - Red) / (NIR + Red + L) (L = 0.5)$
Specific Leaf Area Vegetation Index	$SLAVI = NIR / (Red + SWIR2)$
Simple Ratio Vegetation Index	$RVI = NIR / Red$
Difference Vegetation Index	$DVI = NIR - Red$
Perpendicular Vegetation Index	$PVI = \sqrt{(0.355NIR - 0.149Red)^2 + (0.355Red - 0.852NIR)^2}$
Transformed Vegetation Index	$TNDVI = \sqrt{NDVI + 0.5}$
Renormalized Difference Vegetation Index	$RDVI = (NIR - Red) / \sqrt{NIR + Red}$

Table 3 lists the vegetation indices used in this study.

To meet the spatial independence requirements, a subset of 40% of the total sample points was randomly selected for subsequent modeling and analysis ($n = 1038$). Based on the UAV LiDAR-derived AGB estimation results and severe disturbance monitoring results, this study further pushes the scale of UAV flight area to the scale of satellite remote sensing data. Spectral information, topographic data, stand age, and the mean point cloud height (Hmean) derived from UAV LiDAR were used as independent variables, with the 10 m-resolution sample plot-scale AGB estimation results as the dependent variable. A RF algorithm (numberOfTrees = 100, variablesPerSplit = sqrt, and min_samples_split = 5) was employed concurrently for feature importance assessment and subsequent regional-scale AGB model development. This model then facilitated the estimation of AGB at the regional scale. RF is a machine learning algorithm based on ensemble learning that handles variable interactions and redundancy through random selection of variables and feature sampling. It possesses capabilities for nonlinear relationship modeling, demonstrates resistance to multicollinearity and overfitting, and allows for variable importance assessment [45,46].

To explore the interactive effects and substitution potential of stand age and tree height information in AGB estimation for fast-growing plantations, four distinct modeling scenarios were systematically designed, with the 10 m AGB consistently serving as the response variable. These experiments included establishing spectral data models, stand age–spectral data-enhanced models, tree height–spectral data-enhanced models, and full-variable models. Model performance, quantified by the coefficient of determination (R^2) and Root Mean Squared Error (RMSE), alongside variable importance (%IncMSE), were compared to verify the optimization performance of the models with the introduction of stand age and tree height parameters. The contribution and synergistic effect of stand age and tree height were evaluated.

Subsequently, a regional-scale AGB map for *Eucalyptus* in the study area for 2022 was generated. This was achieved by applying the trained spectral and stand age–spectral models to spectral information from Sentinel data, topography, and stand age data, using the *Eucalyptus* plantation area derived from GF-2 high-resolution tree species classification data as a mask. The optimization effects of incorporating stand age information were systematically analyzed and validated through comparative assessment of spatial distribution between the two approaches.

3. Results

3.1. Stand Age Detection Results

The spatial distribution of disturbances was monitored utilizing the LandTrendr algorithm. Based on using the entire study area as a mask, the spatial distribution of all the detected disturbance times was analyzed (Table 4), with results from Gaofeng Forest Farm shown in Figure 5. Validation and analysis of the stand age identification results indicated an overall accuracy of 80% for this study (Table 5). A detailed examination of the results indicated that certain logging events occurring in 2019 were incorrectly assigned to 2020. This temporal misassignment was attributed to minimal inter-annual variation in the NBR index, leading the LandTrendr algorithm to identify the disturbance in the subsequent year. It is posited that significant cloud contamination prevalent in the study area constrained the availability of cloud-free remote sensing observations, necessitating the use of annual composite imagery for time-series construction. However, if the imagery acquisition occurred earlier in the year and the planting time in the previous year was later, the forest stand would still be in the early growth stage, making it difficult to capture significant differences in the NBR index between consecutive years. This indistinct spectral signature often resulted in the disturbance detection being deferred to the subsequent

year. Nevertheless, based on further analysis and experimental validation, this study concludes that such one-year temporal biases in disturbance identification are considered acceptable for practical applications. Similar temporal discrepancies were also noted in the identification results for 2013 and 2014.

Table 4. Interannual variation in disturbance area across the study area.

Year	2021	2020	2019	2018	2017	2016	2015	2014	2013
Disturbance Area/(ha)	4912.74	12,376.74	2860.47	4082.715	2553.975	2943.72	5539.77	9391.59	2265.48

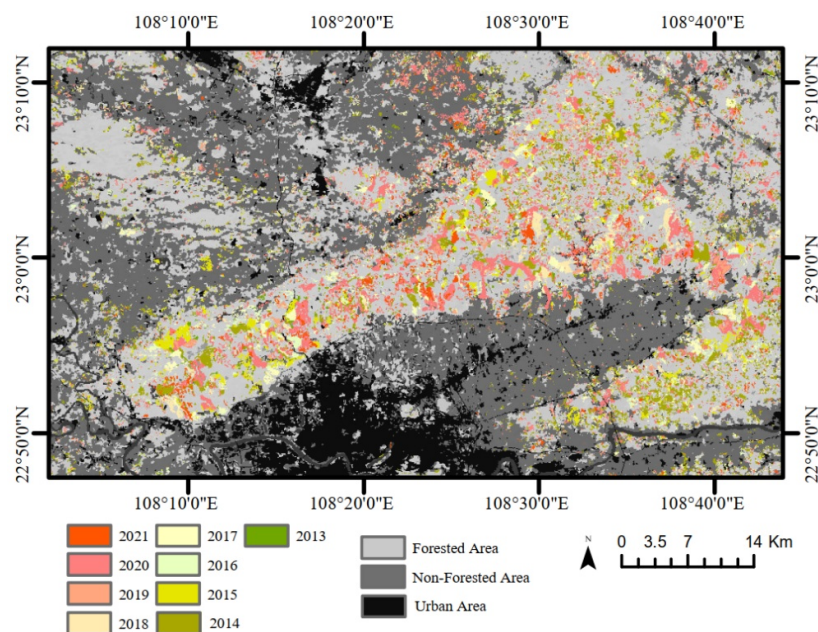


Figure 5. Map of disturbance events identified within Gaofeng Forest Farm.

Table 5. Validation result of plantation disturbance detection algorithms.

	2014	2016	2018	2020	Total
Disturbance Points	37	33	31	39	139
No-Disturbance Points	24	22	23	22	91
Accuracy (%)	85	79	78	83	80

3.2. Regional-Scale AGB Estimation Results

Five-fold cross-validation was employed to evaluate the performance of various RF models and assess the relative contribution of predictor variables (Figure 6a), with the optimal model selected to generate and present the test results (Figure 6b–e). The model based only on spectral information exhibited relatively poor estimation accuracy ($R^2 = 0.54$; RMSE = 31.67 t/ha), with Elevation, SWIR2, SWIR1, B3, and B2 identified as the most influential variables (Figure 7a). Upon incorporating stand age, the model accuracy improved substantially ($R^2 = 0.74$; RMSE = 22.96 t/ha), with stand age emerging as the dominant predictor (Figure 7b). Further improvement was achieved by integrating tree height alone ($R^2 = 0.84$; RMSE = 18.58 t/ha), where tree height superseded other variables in importance (Figure 7c). When both stand age and tree height were included, model performance optimized marginally ($R^2 = 0.85$, RMSE = 17.48 t/ha), suggesting diminishing returns relative to the tree-height-only model. In this combined model, tree height remained the primary explanatory variable, followed by stand age (Figure 7d).

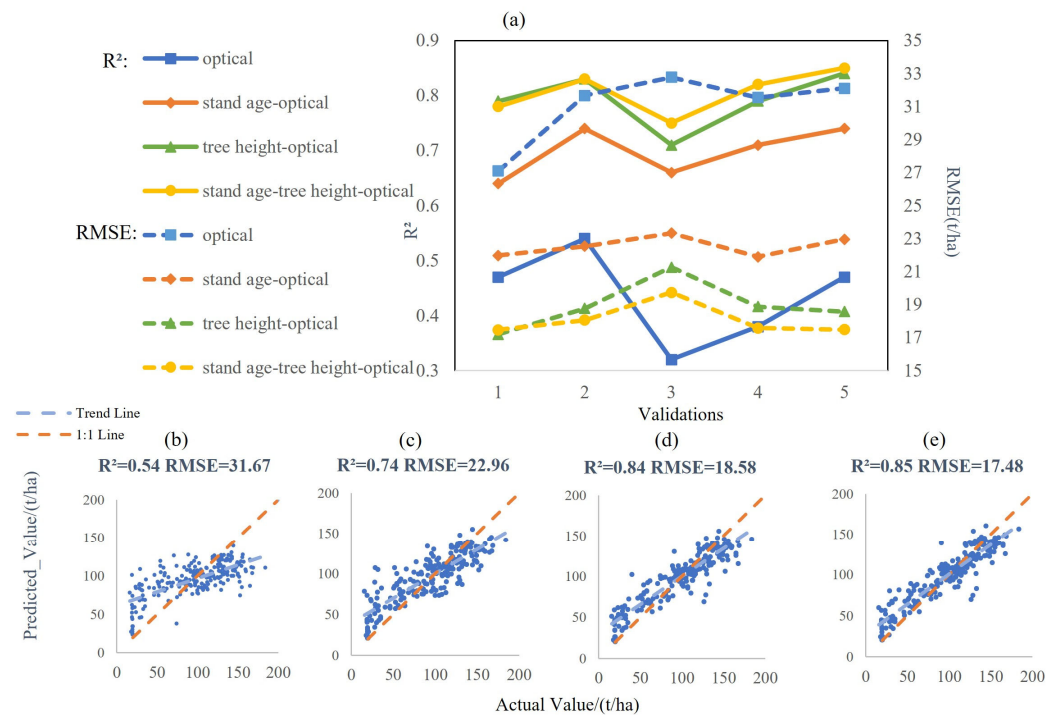


Figure 6. (a) Results of five-fold cross-validation. (b) Prediction accuracy of traditional optical data model. (c) Prediction accuracy of stand age-optical data model. (d) Prediction accuracy of tree height-optical data model. (e) Prediction accuracy of stand age-tree height-optical data model.

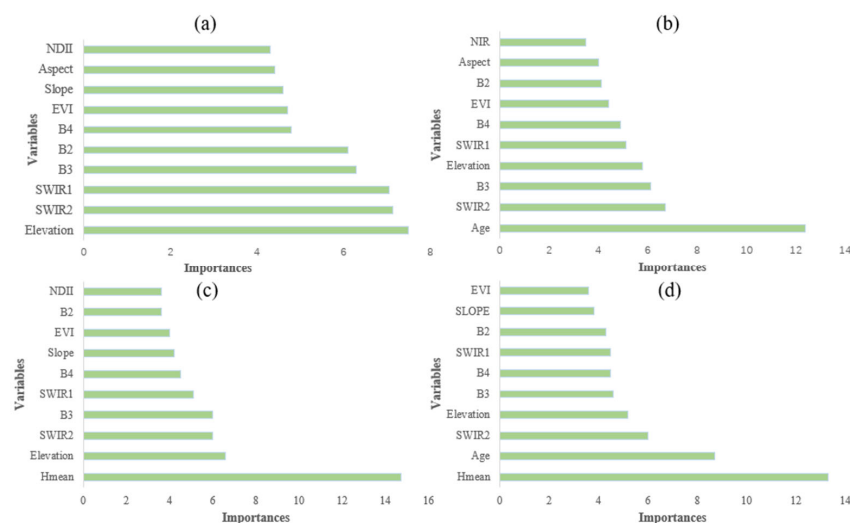


Figure 7. The feature's variable importance ranking for (a) traditional optical data model; (b) stand age-optical data model; (c) tree height-optical data model; (d) stand age-tree height-optical data model.

A detailed examination of scatter plots depicting model performance reveals distinct characteristics across different model configurations. For the spectral model, data points exhibit considerable scatter, with the regression line deviating substantially from the 1:1 line and a slope less than 1 (Figure 6b). Specifically, predictions tend to overestimate actual values in low AGB ranges (e.g., <50 t/ha). Conversely, underestimation is common in high AGB ranges (e.g., >150 t/ha). This pattern is characteristic of the well-documented 'saturation effect' in remote sensing-based estimations, wherein reliance solely on spectral information proves inadequate for accurate AGB quantification, particularly in high-biomass regions where model predictive capability is severely constrained by spectral saturation. Incorporation of stand age (spectral stand age model) led to an improved

overall fit, although some data point dispersion persisted (Figure 6c). The regression slope approached 1 more closely than that of the spectral-only model, and while the tendencies of low-value overestimation and high-value underestimation remained, their magnitudes were attenuated. This suggests that stand age partially mitigates the spectral saturation effect, thereby enhancing model predictive accuracy. The spectral–tree height model exhibited data points more tightly clustered around the 1:1 line, with significantly reduced dispersion (Figure 6d). The regression slope was very close to 1. This indicates that tree height, as a structural parameter, effectively captures critical information correlated with AGB, markedly alleviating spectral saturation limitations and substantially improving model predictive stability and accuracy. The combined spectral–age–height model shows marginal improvement over the spectral–height model alone (Figure 6e). Scatterplot distributions are nearly indistinguishable between these models, suggesting that height likely encapsulates most age-related information about forest growth and structure. This suggests that once tree height, a strong predictor, is incorporated, the additional information contributed by stand age becomes limited. Tree height may already encapsulate substantial information related to forest growth and structure otherwise provided by stand age, resulting in a degree of informational redundancy when both are employed concurrently.

Using the trained RF model in conjunction with detailed tree species classification data, the AGB of the *Eucalyptus* forests across the entire study area was predicted, yielding regional-scale AGB estimation results for the spectral model (Figure 8b). As shown in the statistical histogram (Figure 8a), the average AGB of *Eucalyptus* forests in the study area based on the spectral model is 93.04 t/ha, with the majority of forest area (65.7%) falling within the 90–120 t/ha range. In contrast, low-biomass (<70 t/ha) and high-biomass (>140 t/ha) regions constituted only 17.1% and 0.4% of the total area, respectively. Incorporating stand age as a predictor substantially altered AGB estimates (Figure 8c,d), increasing the mean AGB to 108.13 t/ha. Compared to the model constructed solely using spectral features, the stand age-enhanced model exhibited a statistically significant rightward shift in the AGB distribution, indicating a higher frequency of elevated biomass values. Notably, the proportion of grid cells exceeding 120 t/ha increased markedly, suggesting that stand age data better captures high-biomass canopy conditions where spectral features alone are inadequate.

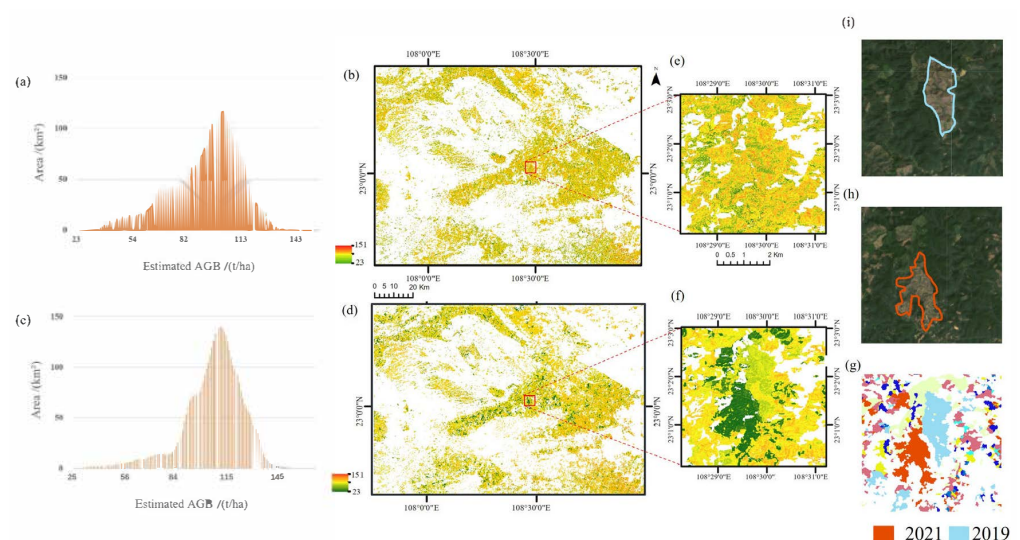


Figure 8. (a) Statistics of spectral model results. (b,e) AGB estimation results from spectral model. (c) Statistics of stand age-spectral model results. (d,f) AGB estimation results from stand age-spectral model. (g) Disturbance detection results. Different colors represent different years of disturbance. (h) Landsat satellite imagery (2021). (i) Landsat satellite imagery (2019).

Moreover, *Eucalyptus* plantations are typically established and managed in extensive, contiguous parcels. In comparison to the spectral information-based estimation model (Figure 8e), the estimation results with stand age information exhibit more coherent block-like spatial distribution features in certain areas (Figure 8f), aligning closely with the temporal planting sequence (Figure 8h–g). Therefore, the current model's estimation results better match the actual spatial distribution characteristics of the plantation forests. This indicates that the model incorporating growth age has a stronger real-world adaptability in capturing the spatial heterogeneity of plantation forests, thereby yielding a more faithful representation of their actual growth dynamics and distribution patterns.

4. Discussions

4.1. Limitations and Biases in Stand Age Estimation

In this study, stand age was derived using the LandTrendr algorithm applied to Landsat time-series imagery, with subsequent verification guided by historical Google Earth imagery. Although this approach demonstrates high efficacy in detecting large-scale, high-severity disturbances (e.g., clear-felling), the resultant age primarily reflects NBR shifts attributable to substantial clear-cut fractions within individual pixels. Consequently, lower-magnitude NBR variations associated with smaller-scale management interventions, such as thinning, selective logging, or replanting, may fall below the detection threshold, leading to their omission [24]. Additionally, due to the use of annual composite imagery to build the time series, if the imagery acquisition time of the previous year was relatively late or if the composite imagery mainly came from early periods of the year, the vegetation may not have fully covered, resulting in a smaller NBR change. Such scenarios can lead to misclassification of disturbance events, manifesting as a temporal lag in disturbance detection and a consequent underestimation of stand age. Moreover, derived stand age was constrained to a maximum of nine years, with stands exceeding this age assigned a uniform value of nine. *Eucalyptus*, a fast-growing plantation species, undergoes rotation cycles typically ranging from 5 to 8 years and shorter cycles as brief as 3 to 4 years [26]. Thus, this upper limit is anticipated to exert minimal influence on overall AGB estimation, particularly as stands older than nine years are infrequent. However, it may introduce an underestimation bias for the limited cohort of older stands. Therefore, in studies of natural forests, it is advisable to refer to Tian et al. (2023) [47], who utilized spaceborne LiDAR data from sources such as GEDI or ICESat-2 to obtain tree height data over larger areas. By leveraging the stand age–tree height relationship, stand age in undisturbed areas can be inferred and subsequently extended, thereby improving the accuracy of certain stand age data.

4.2. Can Stand Age Be Used as a Substitute for Tree Height?

The key premise hinges upon the existence of a significant and stable correlation between stand age and tree height, especially in highly controlled environments like plantation forests. Plantations, particularly the *Eucalyptus* plantations involved in this study, typically exhibit characteristics such as uniform species composition, consistent planting density, and well-regulated growth management practices (e.g., thinning and fertilization) [25,26]. These factors collectively foster a high degree of homogeneity in stand growth and predictable structural characteristics. This ‘clean’ data structure, characterized by minimal confounding disturbances, enables stand age to function as a strong surrogate for tree height, thereby substantially improving AGB model predictive accuracy [27]. However, the efficacy of stand age as a surrogate for tree height requires further quantification in natural forests. Natural forests are characterized by greater species diversity and pronounced growth heterogeneity, leading to significant variability in tree height and biomass, even

among stands of the same age. Complex stand structures (e.g., multi-layered canopies; irregular densities) and the prevalence of natural disturbances (e.g., fire; pests) contribute to substantial spatial heterogeneity, attenuating the age–height correlation [48]. It is worth noting, however, that for dominant species in natural forests, the relationship between stand age and tree height may still exhibit some degree of stability.

Furthermore, even within plantation forests, complex terrain conditions can modulate the stand age–tree height relationship. Terrain factors such as elevation, slope, and aspect directly or indirectly affect site quality, water–thermal conditions, and microenvironments, which lead to differences in growth rates and biomass accumulation patterns, even within monoculture plantations situated across diverse topographical contexts [49]. This study enhances the model’s representativeness by selecting plots with diverse site conditions, but it adopts a unified modeling approach rather than stratifying by terrain, which may affect the generalizability of the stand age–tree height relationship in complex topographies. Moreover, this study was exclusively conducted in highly managed, fast-growing *Eucalyptus* plantations characterized by short rotation periods. The strong correlation observed between forest age and tree height in our findings is likely amplified by the consistent and rapid growth patterns inherent to this specific forest type and its management regime. Our results have not yet been validated in slow-growing species, such as the coniferous forests of North China (e.g., pine and cypress), where the age–height relationship might vary with stand development. However, we postulate that this age–height correlation, which enables stand age to substitute for tree height, is also present in non-fast-growing plantations, at least during their early and mid-growth phases. This relationship likely weakens in later growth stages as height growth plateaus, decoupling it from the linear progression of age. For large-scale AGB monitoring in plantation forests, where the acquisition of detailed structural parameters is often limited, incorporating stand age offers a cost-effective means to substantially enhance model accuracy compared to models relying solely on spectral information. In summary, while stand age demonstrates considerable potential as a surrogate for tree height in relatively homogeneous *Eucalyptus* plantations, its efficacy as a direct substitute in complex natural environments or areas with significant topographical variation requires further validation and quantification. Future research should aim to explore the influence of terrain, stand type, and developmental stage on the age–height relationship and consequent AGB estimation within plantation forests.

4.3. Potential Improvements Following Model Changes

In the preliminary phase of this study, we systematically evaluated conventional regression approaches, including stepwise regression algorithms, for constructing estimation models (Table 6). Initial attempts demonstrated limited predictive accuracy, with the baseline spectral model achieving only modest explanatory power ($R^2 = 0.37$; RMSE = 35.49 t/ha). The inclusion of stand age significantly improved model accuracy ($R^2 = 0.59$; RMSE = 25.97 t/ha), with stand age emerging as the most influential predictor. Further enhancement was observed when tree height was incorporated instead ($R^2 = 0.78$; RMSE = 19.17 t/ha), where tree height became the dominant variable. Notably, the model combining stand age, tree height, and spectral information produced identical accuracy results to the tree height–spectral model, indicating that tree height may already encapsulate much of the explanatory power provided by stand age in AGB estimation within this linear framework—an observation that essentially aligned with the results of the RF model. However, after adding tree height in linear regression, the model and accuracy did not change regardless of whether the stand age was added or not. Analysis of the inter-variable correlation matrix demonstrated a correlation coefficient of 0.762 between stand age and tree height, signifying pronounced collinearity between these two variables. Under

conditions where tree height data were incorporated, the stepwise regression algorithm identified stand age as superfluous information, thereby precluding its integration into the optimal model specification. Given the limitations of linear regression in capturing nonlinear interactions and high-dimensional relationships, this study ultimately adopted the RF algorithm. While the integration of stand age improved overall model accuracy, underestimation persisted in high biomass regions. Other advanced machine learning algorithms such as Support Vector Regression (SVR) and Gradient Boosting Regression Trees (XGBoost) have yet to be explored, which may offer potential avenues to mitigate the ‘regression-to-the-mean’ effect and the attenuation of extreme value predictions often associated with RF outputs [50].

Table 6. Results of the stepwise linear regression models.

Models	Variables and Their Coefficients	R ²	RMSE (t/ha)
optical	AGB = $-6017.440 B3 - 1.452 \text{ Elevation} + 2.508 \text{ Slope} + 3715.787 B2 + 218.591 \text{ NDVI} + 0.055 \text{ Aspect} + 1932.044 B4$	0.37	35.49
stand age–optical	AGB = $10.855 \text{ Age} - 417.754 B11 - 0.709 \text{ Elevation} - 23.033 \text{ TSRI} + 0.147 \text{ Aspect} + 1938.512 B4 - 4940.686 B3 + 4177.704 B2 + 161.781 \text{ DVI}$	0.59	25.97
tree height–optical/stand age–tree height–optical	AGB = $10.375 \text{ Height} - 189.359 \text{ NDII} - 0.413 \text{ Elevation} + 688.231 B2 - 0.509 \text{ Slope}$	0.78	19.17

5. Conclusions

This study, based on the Nanning research area in Guangxi, quantitatively evaluated the individual contributions and combined effects of two pivotal parameters—tree height and stand age—on the accuracy of AGB estimation at the regional scale for fast-growing plantations. This study extracts stand age parameters from time-series remote sensing imagery and integrates tree height data derived from LiDAR to construct a spectral–stand age–tree height collaborative inversion model. Experimental findings revealed that a model predicated solely on spectral features exhibited the lowest predictive performance ($R^2 = 0.54$; RMSE = 31.67 t/ha). The incorporation of stand age yielded a substantial improvement in estimation accuracy ($R^2 = 0.74$; RMSE = 22.96 t/ha), with further enhancement realized upon the integration of tree height ($R^2 = 0.84$; RMSE = 18.58 t/ha). Furthermore, the inclusion of both stand age and tree height parameters did not yield a purely additive improvement in model accuracy ($R^2 = 0.85$; RMSE = 17.48 t/ha), underscoring the predominant influence of tree height in AGB estimation. Although structural parameters, notably tree height, demonstrate high sensitivity and robust predictive capability for AGB estimation in plantation forests, their acquisition frequently relies on resource-intensive and technologically demanding methods like LiDAR, thereby constraining their widespread applicability across extensive plantation areas. In light of the typically strong correlation observed between stand age and tree height within such fast-growing managed systems, employing stand age as a surrogate structural parameter for large-scale AGB monitoring in fast-growing plantations offers a pathway to achieve considerable accuracy improvements while incurring lower data acquisition costs.

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